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MIDDLE CAMBRIAN PALEOMAGNETISM OF THE AVALON TERRANE IN CAPE BRETON ISLAND, NOVA SCOTIA

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Abstract. Interbedded volcanics and sediments of the Bourinot Group (Mid-Cambrian) in central Cape Breton Island were studied as part of a paleomagnetic investigation to quantify the geologically plausible displacements of the Avalon terrane with respect to the North American craton. Both volcanics and sediments reveal prefolding magnetizations with steep to intermediate inclinations. Dual-polarity magnetizations in different volcanic flows are probably primary and they yield a mean direction of D=293°, I=-66° (N=9, k=41.8, α_{95}=8.1°). The sediments reveal, upon detailed thermal demagnetization, complex multivectorial remanences. However, a few samples show characteristic components and a large number of samples (38) have intersecting great-circle remagnetization trajectories; when combined, this yields a mean direction of D=267°, I=-65° (k=16, mean radius of error ellipse is 8.5°). The mean directions for the Bourinot rocks correspond to a paleolatitude of 49°S±11°, which is different from the range of possible values for the predicted Mid-Cambrian paleolatitudes for Nova Scotia (29°S to 18°N), inferred from the North American apparent polar wander path. This discrepancy between predicted paleolatitudes and the observed paleolatitude implies that Nova Scotia was not part of North America during the Mid-Cambrian, but rather, they were separated by an ocean with a minimum width of 1000 km (20°±10°). This interpretation of the relative positions of Nova Scotia and North America based on paleomagnetism is consistent with the geological record in the northern Appalachians and supports previous interpretations of the dissimilar Early Paleozoic faunas in Avalon and North America. It is paleomagnetically quite plausible that the Avalon terrane formed part of a larger continent, comprising Africa (Gondwana) and Armorica during the Cambrian.

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INTRODUCTION

Northern Nova Scotia and Cape Breton Island are part of the Avalon zone, which has attracted renewed attention recently from geologists and geophysicists who are trying to define the extent and nature of involvement of "displaced terranes" in the northern Appalachian fold belt [Keppie, 1982; Lefort and Haworth, 1978; Williams and Hatcher, 1983]. Geologic arguments for the displacement of Avalon are based mainly on indirect evidence, such as the contrasting pre-Silurian geologic records for Avalon and adjacent parts of the northern Appalachians, and the occurrence

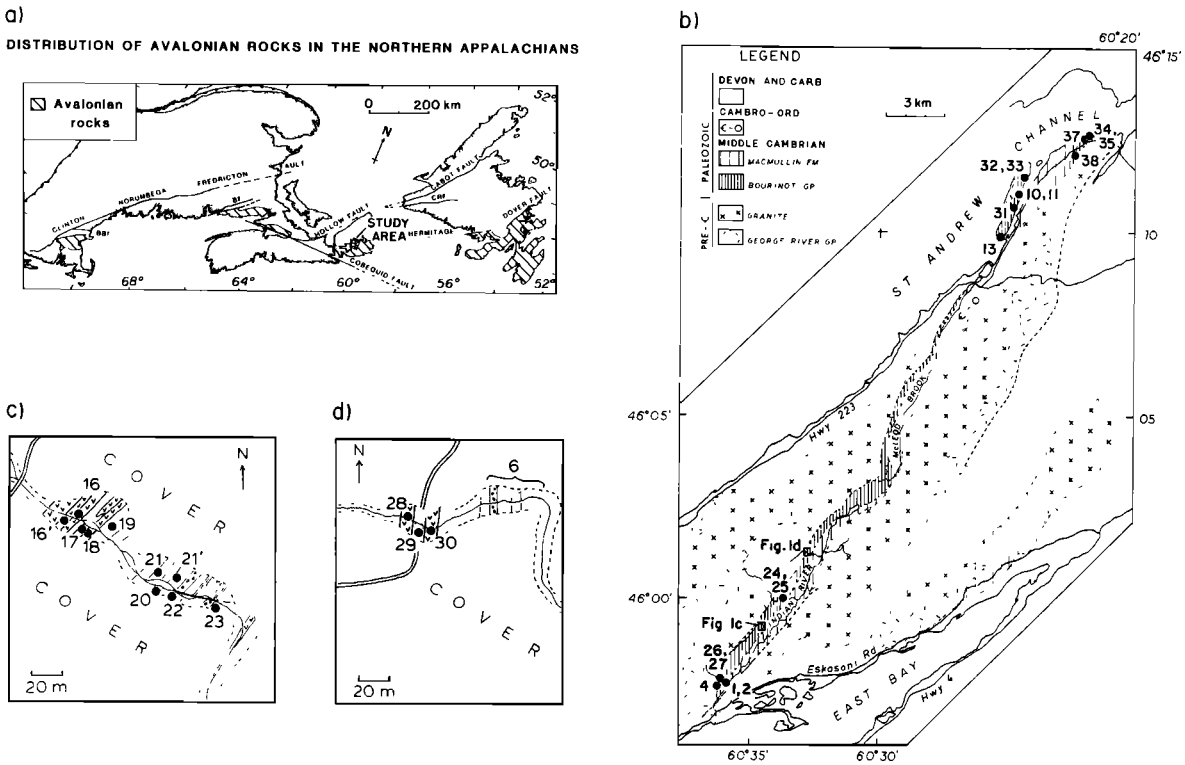


Fig. 1. (a) Regional tectonic map and (b) generalized geological map and paleomagnetic sampling sites in the Boisdale peninsula, central Cape Breton Island [after Hutchinson, 1952]. The location of the study area is indicated in Figure 1a showing the distribution of Avalonian rocks in the northern Appalachians. Map giving detail for sampling localities near (c) Dugald Brook and (d) same for Gillis Brook.

of remnants of possible Paleozoic ocean crust between the Avalon zone and the North American craton in some areas of the fold belt [e.g., Zen and Palmer, 1981; Williams and Hatcher, 1983; Rast and Skehan, 1983]. On the other hand, dissimilar Early Paleozoic faunas between Avalon and cratonic North America have been known for a very long time and these have logically been cited as evidence for prior separation of Avalon and N. America [Wilson, 1966; McKerrow and Ziegler, 1972; Whittington and Hughes, 1972]. More recently, Cocks and Fortey [1982] have examined the distributions of fossil organisms from several phyla throughout the North Atlantic-bordering continents, and by comparing the distribution of species associated with the same paleoecological habitats, they concluded that during the Early Paleozoic there must have been wide latitudinal separation of North America and Maritime Canada across an oceanic barrier. The present paper on Mid-Cambrian paleomagnetic results from Nova Scotia provides a comparison with similar data from cratonic North America as an independent test for the displacement hypothesis.

A few paleomagnetic results from Avalonian rocks in Canada have been published previously. These include early investigations of Late Precambrian strata in Newfoundland [Nairn et al., 1959] and New Brunswick [Black, 1964]. However, these are not considered reliable because demagnetization procedures were either not used [e.g., Nairn et al., 1959] or were inadequate to ensure removal of secondary overprint magnetizations. In a more recent paper, Rao et al. [1981] report a number of widely scattered paleopoles for multiple magnetization components found in Paleozoic age granites from Cape Breton Island. Unfortunately, the ages for individual remanence components in the granites are very poorly constrained; in addition, there is no control on possible

TABLE OF LOWER PALEOZOIC FORMATIONS ON CAPE BRETON ISLAND

Epoch	Paleontological zone	Boisdale Peninsula	Mira area
Lower Ordovician	Tetragrantus	McLeod Brook formation	
Upper Cambrian or Lower Ordovician	Asaphellus-Shumardia		
Upper Cambrian	Peltura	MacNeil formation	MacNeil formation
	Leptoplastus		
	Parabolina spinulosa		
	Olenus		
	Agnostus pisiformis		
Middle Cambrian	Lejopyge		
	Paradoxides forchhammeri		MacLean Brook formation
	Paradoxides davidis		Trout Brook formation
	Paradoxides hicksi	MacMullin formation	
(540 my)	Paradoxides bennetti	● Bourinot group (Gregwa, Dugald, and Eskasoni formations)	
Lower Cambrian	Protolenus		Canoe Brook formation
			MacCodrum formation
	Callavia		Morrison River formation

Fig. 2. Stratigraphic column with a correlation of Lower Paleozoic rocks in Cape Breton Island. The solid circle indicates the strata that were sampled.

postmagnetization tilting of the plutons. This emphasizes the importance of stratified rocks, for which paleohorizontal is known, for paleomagnetic studies within orogenic belts. In another recent study Sparlosu et al. [1983] found probable Late Paleozoic remagnetizations in Late Precambrian sandstones from New Brunswick. Paleomagnetic evidence for the Paleozoic position of Avalon thus has remained inconclusive.

GEOLOGICAL SETTING

The Late Precambrian and Early Paleozoic geological sequence preserved in Cape Breton Island, summarized in Figures 1 and 2, comprises Late Hadrynian basement volcanics and granites overlain by well-preserved Cambrian to Lower Ordovician shallow-water marine clastics and volcanics. The lithologic and faunal characteristics of these rocks and their correlation with similar strata found along strike in the northern Appalachians, particularly in eastern Newfoundland (see Figure 1a), have been discussed by many authors [e.g., Henningsmoen, 1969; Rast et al., 1976; Williams, 1978]. The assemblage of often geographically separate localities where similar rocks occur has been called the Avalon terrane by Williams and Hatcher [1983], named after the Avalon peninsula of Newfoundland.

The Cambrian volcanics present in the Boisdale Peninsula (Bourinot Group) were the principal focus of our study. These volcanics comprise flows and volcanogenic sedimentary rocks, which appear to have had a restricted areal extent at the time they were formed [Keppie and Dostal, 1980; Hutchinson, 1952]. They have been related

by Hutchinson [1952] to a volcanic center located probably in the Boisdale Hills and active during Mid-Cambrian time, whereas little trace of volcanic activity is recorded in a sequence of shales being deposited at the same time less than 50 km away [Weeks, 1954]. The flows are the oldest Paleozoic strata in the Boisdale Peninsula, and they were presumably deposited on top of Hadrynian granites and volcanics occurring there [e.g., Keppie, 1979], although this contact is not exposed. The lowermost flows were deposited subaerially, while the topmost few flows are interbedded with marine sediments bearing Mid-Cambrian trilobites and brachiopods [Hutchinson, 1952]. The amount of time represented by the subaerially deposited flows is uncertain but the upper flows were formed within the lowest (trilobite) stage of the Middle Cambrian (Figure 2) [Hutchinson, 1952]. The composition of the volcanics varies from felsic to basaltic with more felsic flows occurring at the base of the sequence. The total thickness of the volcanics is variable (60-600 m) throughout the Boisdale Peninsula but thickness estimates at many localities are only approximate owing to incomplete exposure; individual flows are from 1 to 5 m thick. According to Hutchinson [1952], the volcanic center in the Boisdale Peninsula was probably an island and it exerted a strong influence on sedimentation in the mid-Cambrian; Mid-Cambrian dark to light-greenish siltstones and sandstones which are interbedded with the flows and overlie them (Dugald Formation) contain a lot of well-worked volcanic detritus.

Lower Ordovician marine black shales in the Boisdale Peninsula are conformable with the Cambrian strata. Following the Lower Ordovician shales there is a gap in the stratigraphic record; the next youngest strata here have an areally restricted occurrence, representing fresh-water arkoses and conglomerates of Mid-Devonian age, and are called the McAdam Lake Formation [Bell and Goranson, 1938] (Figure 1). Although some authors have argued that there is an angular unconformity between the Lower Paleozoic strata and the McAdam Lake Formation [e.g., Helmstedt and Tella, 1973], this unconformity is not exposed and the evidence for it is equivocal. The Lower Paleozoic strata and the McAdam Lake Formation were all tilted steeply about generally northeast-trending strikes before deposition of the overlying Horton Group clastics of Early Carboniferous age. The main phase of folding in this region is therefore bracketed between Middle Devonian and late-Early Carboniferous, suggesting correlation with the Acadian orogeny.

The Bourinot strata form part of a major synclinal structure, a few kilometers wide, which trends the length of the Boisdale Peninsula. In the northern part of the peninsula two opposing limbs of the synclinal structure are preserved, but the northwest-dipping limb is not present farther south. At the southernmost exposure of Bourinot rocks a steep fault paralleling the trend of the syncline juxtaposes basement rocks to the southeast against the southeast-dipping Bourinot strata. The strikes of the Bourinot beds relevant to our sampling are uniformly northeast (mean strike = 35±7°), with the exception of those found on Gillis Brook (Figure 1d) which occur at an irregular bend in the syncline and may be due to local rotations. Minor folds have not been observed by us in the Bourinot strata, but F2 minor folds with an average trend oblique (Azimuth/Plunge = 190/+18) to the major fold axis are present throughout the peninsula in thin-bedded siltstones of the overlying MacMullin Formation. Thus, although a tilt, or plunge of the entire Boisdale syncline as a result of the F2 deformation is unlikely, it may be that differential tilt-rotation of some parts of the syncline has nevertheless occurred. The F2 folds with their observed plunge were probably formed during late-stage dextral shearing of the Boisdale syncline; this mechanism has also been suggested to explain the large fault along the axis of the syncline in the south [Weeks, 1954]. Whereas slip along bedding planes in the sediments may have resulted in a plunge during the F2 deformation, we do not believe that the more massive volcanics along the limbs of the major fold behaved similarly; the strain in the volcanics is likely to have been accommodated instead by strike-slip faulting, involving very little plunge. Axial-planar schistosity is associated with minor folds in the MacMullin Formation, but the Bourinot strata do not show evidence of penetrative deformation. Lastly, and of particular importance to our paleomagnetic study, none of the Paleozoic strata here have been

metamorphosed in temperatures exceeding those of the subgreenschist facies.

SAMPLING

The Bourinot volcanics and sediments are exposed in stream sections in the southern Boisdale peninsula, and in railroad cuts, along the shore of St. Andrews Channel, and on Long Island (see Figure 1, sites 13,31-33) in the north part of the peninsula. Initial reconnaissance sampling in the Bourinot Group was concentrated in sediments of varying lithology exposed in the stream sections along Gregwa, Dugald, Macmullin, and Gillis Brooks. Different lithologies represented here probably reflect varying source materials since fossils preserved in different beds are similar [Hutchinson, 1952]. These streams were later resampled in detail (see Figure 1) with each site restricted to a single bed or a few beds with similar lithology. Volcanic flows immediately underlying or interbedded with sediments were sampled at Dugald and Gillis Brooks (see Figures 1c and 1d), as well as in the north part of the study area. A similar agglomerate unit was sampled at Gregwa Brook (site B01) and at MacMullin Brook (B25) and may represent the same unit [Hutchinson, 1952]; however, other stratigraphic units, whether volcanic or sedimentary, were not uniquely correlated between localities.

The Bourinot Group volcanics and sediments were generally both sampled at a given locality. Although the volcanics are more abundant in some parts of the peninsula, the structural attitudes of these rocks were often difficult to assess where sediments were not present (generally inland). This problem was compounded because local fault-rotations were suspected especially in the basal, exclusively volcanic part of the section near to the contact with basement. As a consequence of this, optimal sampling procedures dictated that throughout most of the peninsula sampling of the volcanics was restricted to a rather narrow stratigraphic horizon, including at most four flows at a given locality (e.g., Dugald Brook, Figure 1c). Even though greater exposure of volcanics on Long Island in the north allows recognition of primary structures in some flows (e.g., compositional banding and contacts between different flows), difficult access to steep and strike-parallel exposures limited sampling here to a couple of flows at separate localities, all within approximately 10 m of the sedimentary part of the section. The sedimentary units of the Bourinot Group sampled upsection from the volcanics were generally within 50 m of plunging minor folds in the MacMullin Formation, indicating that the sampled sediments are likely to have a plunge as well.

All the flows sampled in the south part of the study area are basalts, as were the volcanics sampled at sites B10,35,37 in the north; rhyolites were sampled at three localities in the northern part of the peninsula only (B31,33,38). Site B32 on Long Island was taken in a basalt dike cutting some rhyolites, which was inferred to be a feeder dike for a basalt higher in the section. Almost all of the flows in the southern part of the peninsula show visible oxidation, including flows at B16,16', which were deposited subareally, and B28,30, which were deposited in shallow water. The conditions during deposition of the flow at B17 are less certain because it is underlain and overlain by marine sediments; however, the conspicuous hematite staining in the underlying siltstone, which contrasts with other sediments in the section [Hutchinson, 1952], suggests subareal exposure of the surface on which the overlying flow was deposited. Only the youngest flow occurring at Dugald Brook (site B18; see Figure 1c) was not visibly oxidized, and this may be the result of a faster rate of deposition and burial. Volcanics in the northern part of the peninsula exhibit varying degrees of oxidation, but in general, younger flows in the sequence show better preservation; the flows and dike sampled near the top of the sequence here were relatively fresh. We note that sites B35,37 are in the same thick basalt exposed in two different localities; samples at B35 all come from within a few centimeters of its base whereas the other site was taken in the interior of the flow.

Sampling in the Boisdale Peninsula was done with a portable gas-power-ed drill and diamond core-bit, insofar as possible. Samples thus collected were oriented with a Brunton compass and inclinometer. Block samples, also oriented with magnetic compass, were taken in the volcanics on Long Island where access with a drill was particularly difficult. The effect of the remanence of

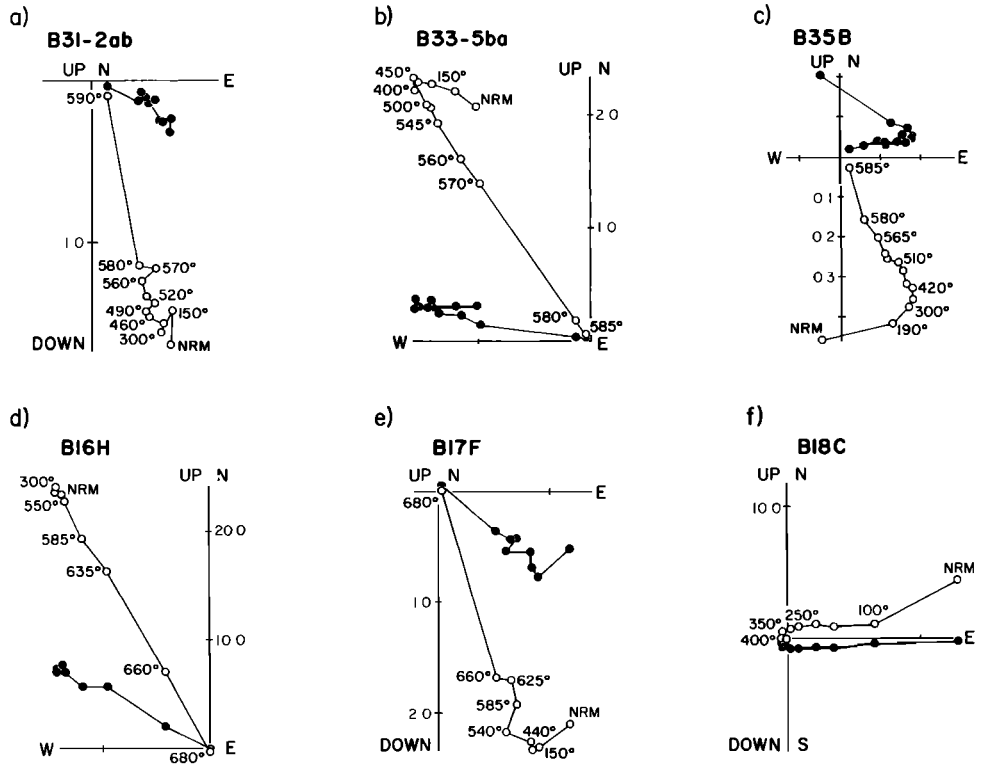


Fig. 3. Orthogonal demagnetization diagrams [Zijderveld, 1967] showing stepwise thermal demagnetization of Bourinot volcanics. The first two digits of the sample numbers indicate the site (see Figure 1 and Tables 1 and 2). The filled symbols represent projections of the remanence vector onto the horizontal plane, and the open symbols represent projections onto an east-west vertical plane. Labels for open symbols indicate the temperatures (C) for treatment steps. The axes are labeled in units of 10^{-2} A/m. Directions for specimens have been corrected for tilt.

the volcanics on the Brunton measurements was deemed to be insignificant because of the low NRM intensities (10^{-3} to 10^{-1} A/m). Estimates of paleohorizontal at each sampling locality were obtained from the average of at least three measurements per locality.

PALEOMAGNETIC RESULTS

Paleomagnetic specimens, prepared from oriented core and block samples, were demagnetized using standard alternating field (AF) and thermal methods. The results were analyzed using orthogonal projection diagrams [Zijderveld, 1967]. Initial AF and thermal demagnetization results for specimens from the same rock unit showed that magnetization components resolved by thermal treatment were not separated using AF. The majority of the remaining specimens were therefore thermally demagnetized. Demagnetization data for specimens not yielding stable endpoints in thermal treatment were further examined for coplanarity of successive vector endpoints, using the remagnetization-circle method of Halls [1976,1978] and Bailey and Halls [1984].

Bourinot Volcanics

Intensities for natural remanent magnetization (NRM) measured in the Bourinot volcanics varied widely between sites. Thermal demagnetization behavior of volcanics also varied between sites, but this behavior was strongly dependent on the presence or absence of visible oxidation products in the volcanics, as also observed in the field. Within-site agreement in laboratory unblocking-temperatures (Tub) and directions for components removed during demagnetization

TABLE 1. Site Mean Paleomagnetic Data for the Bourinot Group Volcanics

Site Number	Latitude(N),Longitude(W) ,deg	Bedding Strike/ Dip, deg	n/n_o	Before T.C. Decl/Incl	After T.C. Decl/Incl	k_2	α_{95}	k_2/k_1	Magnetic Carrier
10	46.18 , 60.41	37/67	7/7	136/-35	287/-73	101	6.0		mte
13	46.16 , 60.42	47/46	0/6	(components not separated)					
16	45.99 , 60.58	44/83	9/9	143/-28	293/-67	66	6.4		hte
16'	45.99 , 60.58	44/83	3/3	149/-27	280/-65	121	11.2		hte
17	45.99 , 60.58	44/83	7/7	313/+31	136/+67	49	8.7		hte
18	45.99 , 60.58	44/83	0/4	(components not separated)					
28	46.02 , 60.55	356/83	0/6	(components not separated)					
30*	46.02 , 60.55	356/83	5/5	[140/+00]*	[169/-34]*	90	8.1		hte, mte
31	46.18 , 60.41	37/67	4/4	304/+42	133/+70	84	10.1		mte
32	46.19 , 60.40	18/49	4/9	59/-68	316/-52	33	16.2		mte
33	46.19 , 60.40	18/49	3/3	111/-80	288/-51	275	7.0		mte
35	46.21 , 60.37	213/73	8/10	110/+09	64/+75	35	9.2		mte
37	46.21 , 60.37	213/73	0/7	(see Table 2, and text)					
38†	46.20 , 60.37	218/73	4/4	294/-02†	263/-70†	84	12.0		mte
Mean			9/14		293/-66	42	8.1	18.1	

Site numbers correspond to those of Figure 1. n/n_o is the ratio of samples used in the statistical analysis to those thermally demagnetized. Decl/Incl are declination and inclination, in degrees; T.C. is correction for simple tilt of the strata;k and α_{95} are the statistical parameters associated with the mean direction [Fisher, 1953], k_1/k_2 is the ratio of the precision parameters after and before the correction for the tilt of the strata, and for these rocks represent a statistically significant fold test; the magnetic carrier is discussed in the text and is either magnetite (mte) or hematite (hte).

*Magnetization at this site was anomalous and has not been used in the mean direction.

†The mean direction for this site was determined from the intersection of remagnetization circles; for this site is the mean radius of the 95% error ellipse about the intersection [Bailey and Halls, 1984].

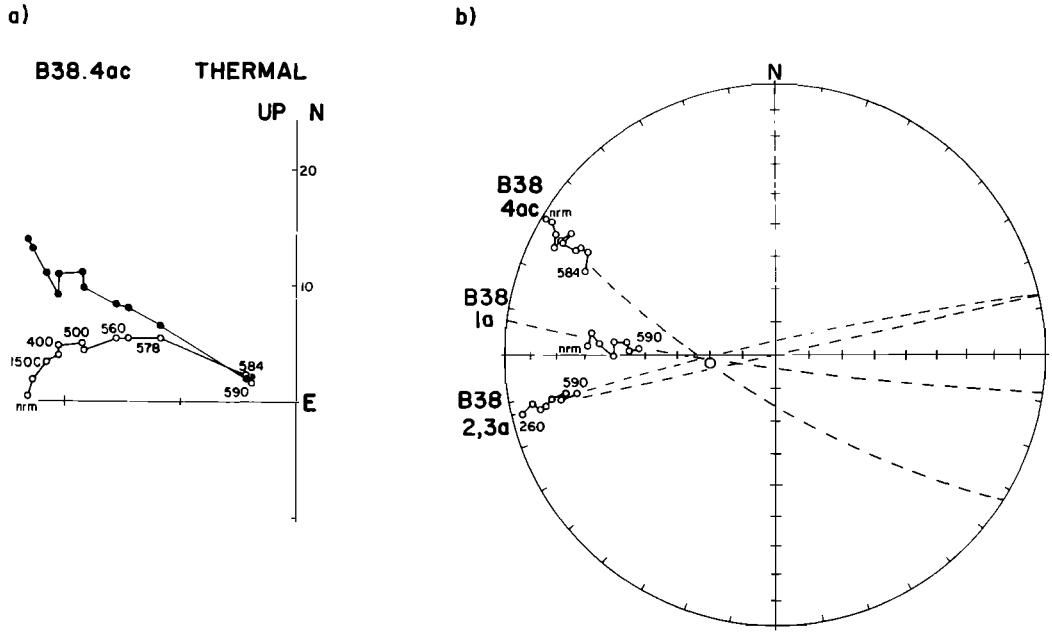


Fig. 4. (a) Orthogonal demagnetization diagram illustrating thermal treatment of a specimen from site B38, using same conventions as Figure 3. (b) Equal-area projection showing successive magnetic directions (tilt-corrected) during thermal demagnetization for several samples from site B38. Open (closed) symbols indicate upper (lower) hemisphere. Dashed lines indicate upper hemisphere remagnetization circles calculated for each specimen, and the large open symbol corresponds to the best-fit intersection of sample remagnetization circles [Halls, 1976].

treatment in both oxidized and unoxidized flows was very good.

The unoxidized volcanics, in general, revealed multivectorial magnetizations. Most showed a component of magnetization removed in thermal demagnetization up to about 300°C (e.g., Figures 3b and 3c). Since this component, when not corrected for tilt (in situ), has northerly declinations and downward inclinations, this is probably a viscous magnetization (VRM) acquired in the present earth's field. After removal of the low-Tub component, magnetization in specimens changed only slightly in continued thermal treatment until, at temperatures greater than 550°C, the magnetization decayed univectorially to the origin with complete removal of remanence by 590°C (Fig. 3a-3c). The directions for these last-removed components of magnetization are consistent at the site level and they are the characteristic directions for which site-means are reported in Table 1 for sites B10,31,32,33, and 35. In samples from one other site in this group (B38), magnetization directions changed continuously during demagnetization treatment and stable end-points were apparently not achieved (e.g., Figure 4a); yet, magnetization directions for several samples converged toward the characteristic direction with progressive treatment. These samples from B38 revealed different great-circle trajectories for successive directions measured during treatment, with a mutual point of intersection of great-circles (westerly and steeply up; see Figure 4) This is identified as the characteristic direction in common to samples from this site (Table 1). Individual remagnetization circles and their intersection were computed using the Halls [1976] method.

Discrete and fairly uniform maximum blocking temperatures for the unoxidized volcanics are clearly indicated in Figure 5, showing intensity of NRM vs. temperature for representative volcanic samples. The shape of the curves vary, owing mainly to the direction of low-Tub

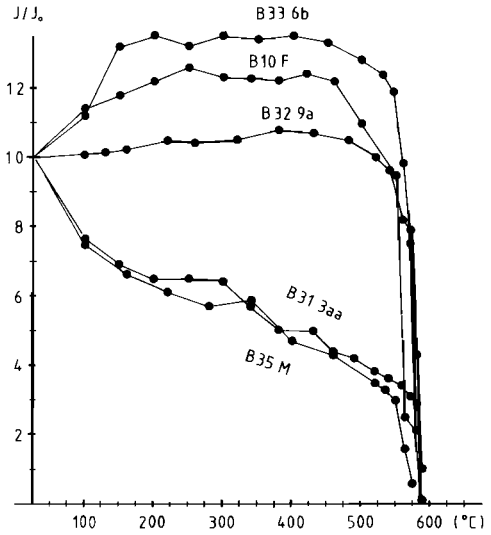


Fig. 5. Thermal demagnetization of natural remanent magnetization showing normalized intensity (J/J_0) in unoxidized volcanics with Curie temperatures typical of magnetite.

components relative to the characteristic direction at different sites; the low-Tub direction clearly opposes characteristic directions at some sites. Discrete Tub in the range of the Curie-temperature of relatively pure magnetite strongly suggests that it is the mineral carrier of remanence observed in these units.

A second group of volcanic sites includes the oxidized flows sampled in the southern part of the field area. Specimens from these flows revealed very stable univectorial magnetizations with discrete Tub of 660°-680°C (e.g., Figures 3d,3e). The very high Tub components at these sites are clearly carried in hematite, yet the remanence directions observed for the oxidized flows are similar to those found for the unoxidized flows (compare Figure 3). Two separate flow units in this region, sites Bl6 and Bl6', had nearly identical directions, whereas directions in a third slightly younger flow (site Bl7) were opposite to those found at Bl6,16' (compare Figures 3d and 3e). Directions for high-Tub components in each flow are consistent and they are the characteristic directions reported for these sites in Table 1. Several samples from one additional flow in this southern region (B30) also exhibit high-Tub (>590°C) univectorial magnetizations. However, directions found in these samples are different from any of those observed either in flows or in sediments throughout the sampling area. Remaining samples from this site are of interest because they displayed lower Tub magnetizations (less than 590°C) probably carried in magnetite with the same direction as that carried by hematite at this site. A mean for site B30 is entered in brackets in Table 1 and has not been included in the overall mean.

Other volcanic units that were studied and in which the typical characteristic magnetizations were not found include sites Bl3,18,28, and 37. Although samples from site B37 were obtained from the same flow unit as B35, they do not exhibit the same remanence properties. After removal of a "soft" component in the direction of the present earth's field, samples from B37 yielded directions with declination to the south and shallow inclinations. In a later section, southerly and shallow directions found in sedimentary samples from the Bourinot Group are interpreted to represent a secondary (Late Paleozoic) overprint magnetization (pTRM), and the observed remanence at B37 probably had a similar origin. Magnetization in specimens from site B28 continued to change direction in treatment above 590°C; however, acquisition of laboratory viscous magnetizations in these samples, probably resulting from mineralogical change, prevented resolution of higher temperature components where they were present. Sites Bl3 and Bl8 were dominated by a component with low to intermediate Tub (e.g., Figure 3f). Curved demagnetization trajectories for specimens from these sites indicate the presence of other components, but they could not be resolved accurately.

Age of Characteristic Magnetization in the Volcanics

The nine sites in the Bourinot volcanics which yielded characteristic magnetizations are from different limbs of the Boisdale syncline (see Table 1). Site-mean directions for these sites, given in Table 1, are also shown in equal-area plots, before and after simple tilt correction, along with sample directions (Figure 6). In situ site means for the volcanics are distributed in a girdle trending approximately perpendicular to the strike of the Bourinot strata. Tilt-corrected site means (Figure 6), on the other hand, are either to the WNW and

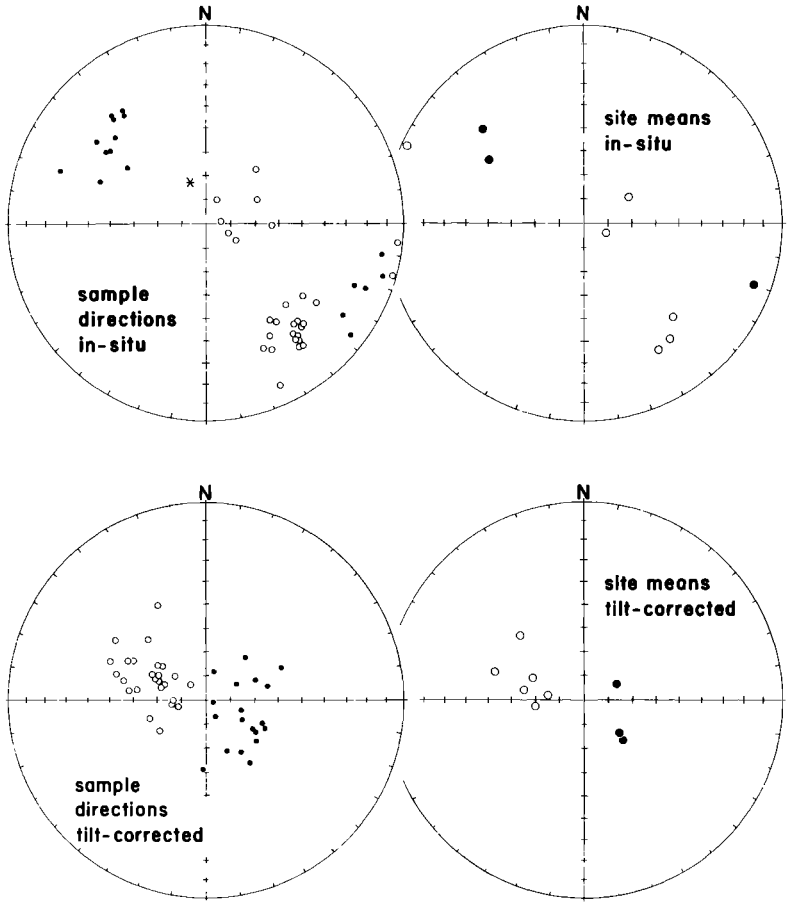


Figure 6. Equal-area projections of sample directions and site-means for the Bourinot volcanics both in situ and after simple tilt correction (Table 1). Open symbol with a dot denotes the site mean from B38 determined from the intersection of remagnetization circles; other symbols are the same as for Figure 4.

steeply up or ESE and steeply down, resulting in two groups of directions nearly oppositely directed, and providing a positive fold test. The statistical parameters for this test (Table 1)[McElhinny, 1964] indicate that it is significant at the 99% confidence level. We particularly note that the tilt-corrected directions for the volcanics do not show large, or systematic variations in declinations indicative of significant undetected plunge of these units. This confirms our previously noted observation about the volcanics, in contrast to the situation of the sediments discussed in a later section. The use of dual-polarity magnetizations in the fold test is justified because the opposite polarity magnetizations indicated in Figure 6 were observed in separate flows on different limbs of the fold. Both polarities were observed in separate hematite-bearing (oxidized) flows, as well as in separate magnetite-bearing (unoxidized) flows. The similar directions in oxidized and unoxidized flows imply a similar age for their magnetizations.

The positive fold test indicates that magnetizations in both groups of volcanics were acquired before tilting of the Bourinot strata; moreover, the presence of dual-polarity magnetizations in separate magnetite-bearing flows places additional constraints on the age for magnetization in these flows, and they strongly suggest that the magnetizations are primary. This is because remagnetization of magnetite in these flows, by either thermal or

thermochemical processes, such that a given flow possesses exclusively one polarity or the other, is unlikely. The only plausible explanation is that different polarity magnetizations were acquired as primary TRM during cooling at slightly different times, in either a normal or reverse-polarity ambient field.

By contrast, the age for the characteristic remanence in the oxidized flows depends upon independent evidence for the origin of the hematite in these flows. On the one hand, it could be argued that because only the lower part of the sampled section is oxidized, the formation of hematite must have been early, prior to deposition of overlying unoxidized flows and sediments. We note that early oxidation is also consistent with field evidence for subaerial deposition of several of these flows. Early oxidation is expected, for instance, in the case of sites B16,16' and probably B17 (see Figure 1c). The presence of a uniform-polarity remanence in each flow, while separate flows have different polarities, also favors early and presumably fairly rapid oxidation of individual flows, because these would be difficult to produce by later remagnetization. We therefore infer that the remanence carried by hematite in the oxidized flows was acquired at the same time, or very soon after the formation of the flows.

Characteristic magnetizations in both groups of flows in Table 1 thus appear to be Cambrian-age magnetizations, although they were acquired by different processes. Directions for characteristic magnetizations in the volcanics, shown after correction for tilt in Figure 6, are therefore inferred to record the paleomagnetic field direction relative to Nova Scotia in the Mid-Cambrian.

Bourinot Sediments

Samples of the sedimentary units of the Bourinot Group exhibited maximum unblocking temperatures (Tub) generally in the range observed for the volcanics: greenish and dark-gray to maroon colored siltstones, which represented the majority of samples, had maximum Tub of about 590°C, while a small number of samples from reddish units yielded maximum Tub up to 660°C. In nearly all siltstone samples, however, unblocking temperatures were continuously distributed, and stepwise demagnetization of these sediments revealed complex multivectorial magnetizations. There was, in addition, considerable within-site variability in Tub for removed magnetization components in the siltstones. A single site (B29) in a sandstone unit with very stable univectorial magnetizations provided an important exception to this pattern of behavior.

Typical demagnetization diagrams for samples from the southern part of the sampling area are shown in Figure 7a-f. A soft component approximately in the direction of the present-day field (PDF) was removed first in all samples. In subsequent thermal treatment most siltstone samples showed a component with southerly and shallow in situ directions removed over an intermediate range of laboratory temperatures (300°C to ~500°C; Figures 7a-c). In some samples this southerly and shallow direction was the only high-temperature component (Figure 7a), but in general, demagnetization trajectories were more complex and revealed the presence of additional components of magnetization with Tub generally higher than those for the intermediate south and shallow component (Figures 7b,7c,7e). Curved demagnetization trajectories resulting from incomplete separation of multiple magnetization components were apparent for many of the Bourinot sediments (e.g., Figures 7b-7e). The curved demagnetization trajectory below 450°C in B22.G (Figure 7d) suggests the presence of both PDF and south and shallow components but neither was isolated. We note that the apparent intermediate component in B21'.Da with southerly declination and intermediate inclination was seen infrequently and probably represents a composite of multiple "real" magnetization components.

Directions for southerly components were found by vector subtraction in 52 of the Bourinot samples, and site means for these are given in Table 2 and Figure 8. This component is generally not seen in the northern part of the sampling area except in the volcanic unit sampled at site B37; the south and shallow component found in sample B37.F is shown in Figure 7g and a mean direction for this site is also given in Table 2. The fold test for the southerly component is clearly negative (Table 2) indicating a postfolding age. The overall mean

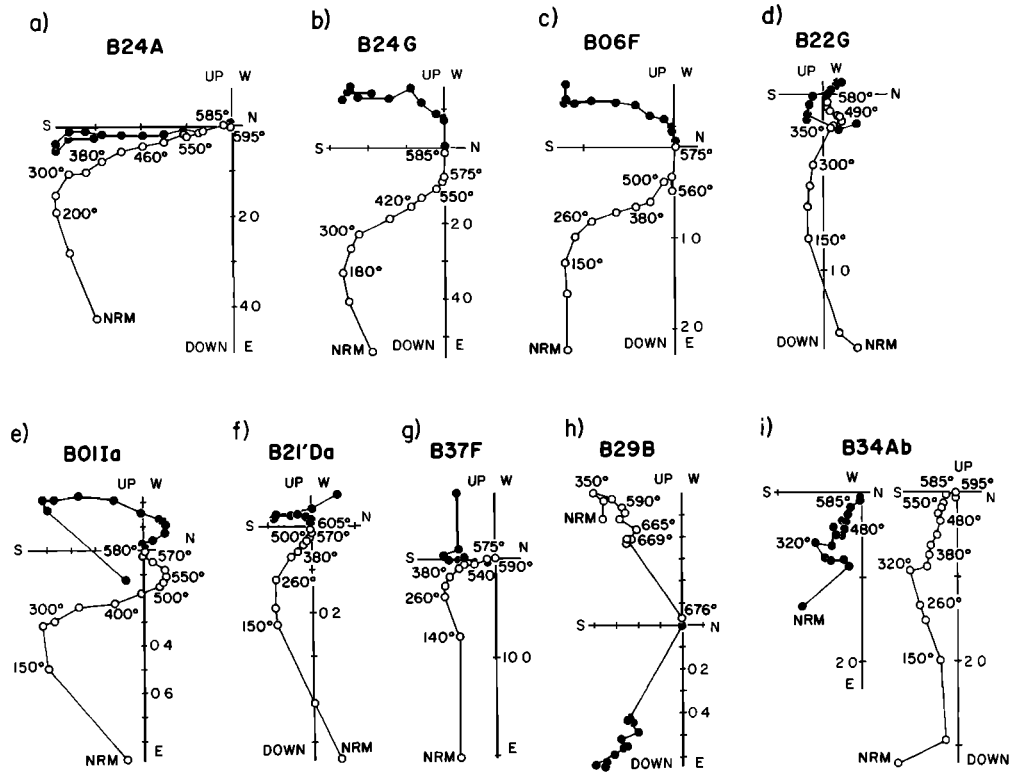


Fig. 7. Orthogonal demagnetization diagrams showing thermal demagnetization of Bourinot sediments and one similarly behaved specimen from volcanic site B37 (Figure 7g). The symbols and units are the same as for Figure 3.

direction for this component, its observed polarity, and its corresponding paleopole (Table 3), are similar to those found in Late Paleozoic strata throughout the Appalachians [Roy et al., 1983; Roy and Morris, 1983] which are thought to have been acquired during the Kiaman reversed polarity interval.

An attempt was also made to analyze high-Tub components in the Bourinot sediments. However, continuously curved high-temperature demagnetization trajectories (e.g., Figure 7c) in the majority of specimens prevented resolution of these components by simple vector subtraction or stable end points. High-Tub components of magnetization appear to be fully defined and isolated in only a few of the Bourinot sediments. Figures 7d and 7e, for instance, show a northwesterly direction remaining at 450°C which was removed in subsequent thermal treatment. Although by comparison with the behavior of other samples (e.g., Figures 7b and 7c) these components seem to be more clearly defined, it is only from additional arguments, as discussed below, that we deduce these components to be relatively free of overprints. One such argument comes from an example of a univectoral magnetization (albeit of polarity opposite to those of Figures 7d and 7e) in a sandstone sample from site B29, as shown in Figure 7h. Other samples from this site showed similar directions despite having been partially baked by an overlying flow (B30), and all samples from this site revealed stable magnetizations with discrete Tub near 675°C. After simple tilt correction, the site mean for B29 (Figure 9a) has a similar inclination to that found for the volcanics but the declination for this site differs somewhat from that found for the volcanics. Since both the strike of the beds and the declination are offset by a similar counterclockwise amount with respect to those for other sites a local structural rotation is suggested. If rotation occurred it must have been before the time of acquisition of the Kiaman overprint, because the southerly and shallow

TABLE 2. Paleomagnetic Data for the Intermediate Blocking Temperature Component

Site Number	Latitude(N), Longitude(W), deg	Bedding Strike/ Dip, deg	n/n_o	Before T.C. Decl/Incl	After T.C. Decl/Incl	k_1	α_{95}	k_1/k_2
01	45.96 , 60.60	32/105	10/14	178/+05	216/-34	56	6.5	
02	45.96 , 60.60	32/105	6/8	177/+07	214/-35	56	9.0	
04	45.96 , 60.60	18/96	1/8	176/+17	182/-23	•••	•••	
06	46.02 , 60.55	356/83	4/6	174/+17	158/+00	41	14.5	
11	46.18 , 60.41	37/67	0/10	(components not separated)				
19	45.99 , 60.58	44/83	0/3	(components not separated)				
20	45.99 , 60.58	44/83	0/2	(components not separated)				
21	45.99 , 60.58	44/83	2/6	175/+15	195/-43	155	20.2	
21'	45.99 , 60.58	44/83	1/3	174/+21	188/-41	•••	•••	
22	45.99 , 60.58	44/83	0/3	(components not separated)				
23	45.99 , 60.58	44/83	3/4	172/+20	187/-44	61	15.9	
24	46.00 , 60.56	37/70	5/5	182/+19	184/-23	69	8.1	
25	46.00 , 60.56	37/70	7/11	173/+09	188/-36	26	12.2	
26	45.96 , 60.60	32/105	4/4	174/+07	215/-38	98	9.3	
27	45.96 , 60.60	32/105	4/4	181/+09	210/-32	60	11.1	
34	46.21 , 60.37	213/73	0/7	(components not separated)				
37*	46.21 , 60.37	213/73	5/7	173/+17	220/+42	61	15.9	
Mean			12/17	176/+14	196/-28	154	3.5	18.5

For explanation of symbols and abbreviations, see Table 1.
*Volcanic flow with remanence characteristics similar to those found in the sediments.

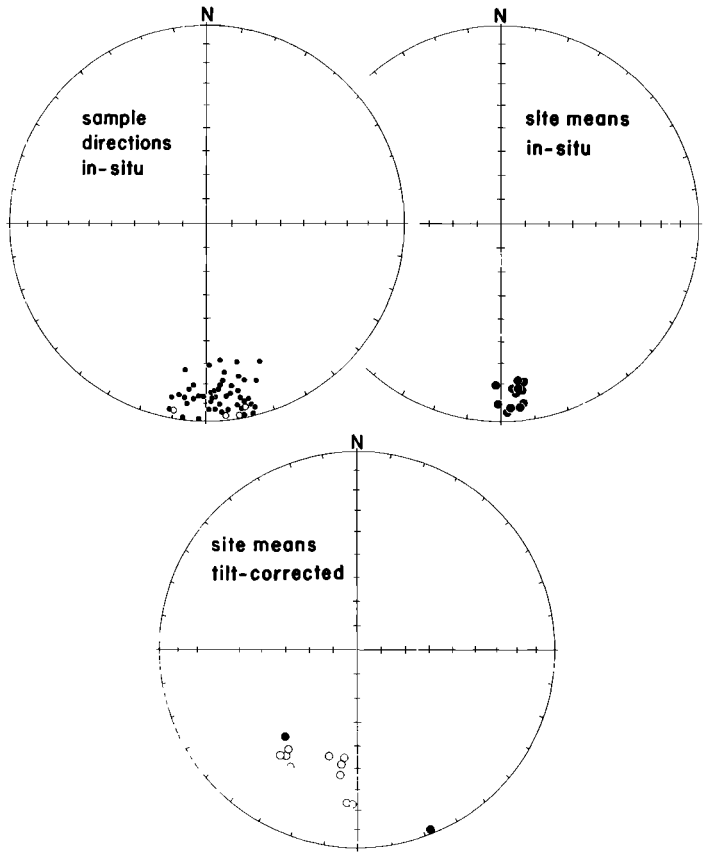


Fig. 8. Equal-area projections of in situ sample directions, and site mean directions both in situ and tilt-corrected, for intermediate unblocking-temperature components in the Bourinot sediments and volcanics (Table 2). Symbols are the same as for Figure 4.

components in overlying strata at this locality do not show declination deviations (compare Table 2, site B06). A strike correction [Perroud, 1983] to the direction for B29 yields the directions with the carat symbol of Figure 9b, shown along with sample directions for high-Tub components in other sedimentary sites which have been corrected for the plunge of nearby minor folds as well as the tilt of the strata. The resultant distribution of sample directions for these sediments, although few in number, is similar to that observed for the volcanics and they yield a similar mean (Table 3).

Although high-Tub components are actually isolated in only a small number of the Bourinot sediments (Figure 9), Zijderveld diagrams and equal-area plots of successive directions during progressive thermal demagnetization suggest that similar high-Tub components may be present in a much larger number of the sediment samples. Equal-area plots for several specimens including B24.G, B06.F, B21'.Da, and B34.Ab, are illustrated in Figure 10. The orthogonal demagnetization diagrams for these samples (e.g., Figures 7b, 7c, 7f, and 7i) clearly show that the remanence directions changed continuously during thermal demagnetization. However, above about 300°C the successive remanence directions for each specimen define a great circle trajectory. The high-Tub portion of the remanence in each of these samples is therefore interpreted to be a sum of two (and not more than two) superimposed components of magnetization with different directions; the plane in common to both, when projected on an equal-area plot, gives the remagnetization circle for that sample [Halls, 1976]. In situ great circle trajectories in Figure 10a show

TABLE 3. Paleopoles From This Study

Magnetic Component	Mean Decl/Incl	Paleopole Location	dp, dm	ϕ
Volcanics, high temperature	293/-66	21S, 160E	11, 13	49
Sediments, high temperature				
Subtracted vectors	278/-59*	24S, 176E	13, 17	40
Remagnetization circle intersection	257/-66	39S, 177E	15, 19	49
Combined remagnetization circles and subtracted vectors	267/-65*	33S, 174E	11, 13	47
Secondary overprint (Table 2)	176/+14	37N, 126E	2, 4	7

Decl/Incl are declination and inclination; dp and dm are the semiaxes of the oval of 95% confidence associated with the pole; ϕ is the paleolatitude of the sampling site; all numerical values have units of degrees.

*All directions for the sediments have been obtained by using a two-stage tilt correction, incorporating the strike correction (site B29) and plunge correction.

that directions in samples from the southern part of the field area move away from a southerly (and shallow) direction with progressively higher temperature treatment. These in situ great circle paths tend toward northwesterly directions, but the arrows indicate that great circle paths for individual samples first converge, and then continue to diverge at high temperature. The southerly and shallow component is not apparent in samples from site B34 in the northeastern part of the syncline (see Figure 7i). Magnetization in samples from B34 was dominated by a northerly and downward direction which was removed first in thermal treatment; with progressive demagnetization the sample directions consistently moved along great circle paths toward an east-southeasterly and shallow direction (different from the south and shallow component observed at B37) but no stable end points were achieved. In the great circle path for sample B34.Ab plotted in Figure 10a, the direction clearly moves away from (and not toward) in situ great circle paths for the other samples shown in the figure.

However, after tilt correction (Figure 10b) the great circle paths for all samples converge toward an easterly and intermediate to steep downward direction. This consistent trend of sample great circles suggests the occurrence of a high-Tub prefolding component of magnetization in common to all these siltstone samples, and the approximate intersection of sample great circles gives an estimate of the direction for the high-Tub component (in tilt-corrected coordinates) although it cannot be isolated in individual samples.

This analysis was extended to a total of 38 of the Bourinot sediment samples, including the ones just described. In the above example and in the analysis of remaining sediment samples, "best-fit" great circle paths for each sample were computed using Halls' [1976] method. In fitting great circles to sample data we used Zijderveld diagrams for each sample to infer the beginning of a great circle path following removal of the third soft (PDF) component of magnetization. The errors associated with individual sample great circles, estimated from the mean radius of the error ellipse about the pole

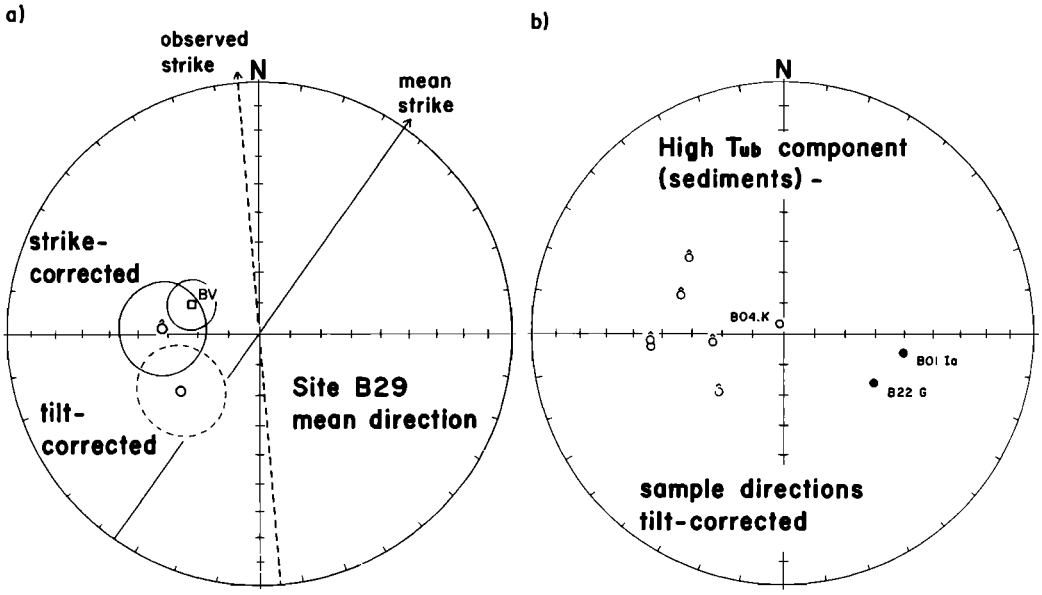


Fig. 9. (a) Equal-area projection of the site-mean direction and circle of 95% confidence for site B29 after simple tilt-correction (D/I = 234°/-58°, k=22, α_{95}=14.6°) and after strike-correction (with carat symbol; D/I = 273°/-58°)[Perroud, 1983]. The observed strike was measured at Gillis Brook; the mean (reference) strike is the mean for all other localities. The overall mean for the characteristic magnetization in the volcanics (BV) is shown for comparison. (b) Sample directions for high-Tub components in the Bourinot sediments. Sample directions from site B29 (carat symbol) have been strike corrected; the other directions have been corrected for plunge as well as tilt of the strata. Conventions are the same as for Figure 4.

(Bailey and Halls, 1984), did not vary greatly between samples (4°-12.6°) and we did not reject any data on the basis of these errors. Poles to acceptable remagnetization circles thus calculated for 38 of the Bourinot sediment samples are shown in Figure 10b. An east-west girdle of in situ remagnetization circle poles in Figure 10b reflects the presence of a south and shallow component in many samples. After tilt correction (again a two-stage correction was used) the poles to remagnetization circles define an approximate great-circle girdle representing a subhorizontal plane (slightly inclined toward the west; Figure 10d). Thus, this set of samples has, when tilt-corrected, a second common intersection, with an easterly and steeply downward direction (perpendicular to the subhorizontal plane containing poles to remagnetization circles). Convergence of tilt-corrected great circle trajectories consistently toward this intersection point supports the interpretation of a high-Tub component in common to this set of sediment samples. Because the high-Tub magnetization has a common direction in different samples or sites only after correction for tilt, this magnetization is inferred to have been acquired before folding of the Bourinot strata. Following Halls' [1976, 1978] method, a best-fit great circle to the tilt-corrected remagnetization circle poles yields a best estimate of D=77°, I=+66° (solid square in Figure 10b), for the intersection (Table 3).

The algorithm of Bailey and Halls [1984], used to estimate errors associated with best-fit great circles, was also applied to the sample remagnetization circle poles of Figure 10b to estimate the precision parameter and error associated with the great circle intersection; their method, moreover, enabled us to combine the great circle and stable end point data of Figures 9 and 10b to give a sediment

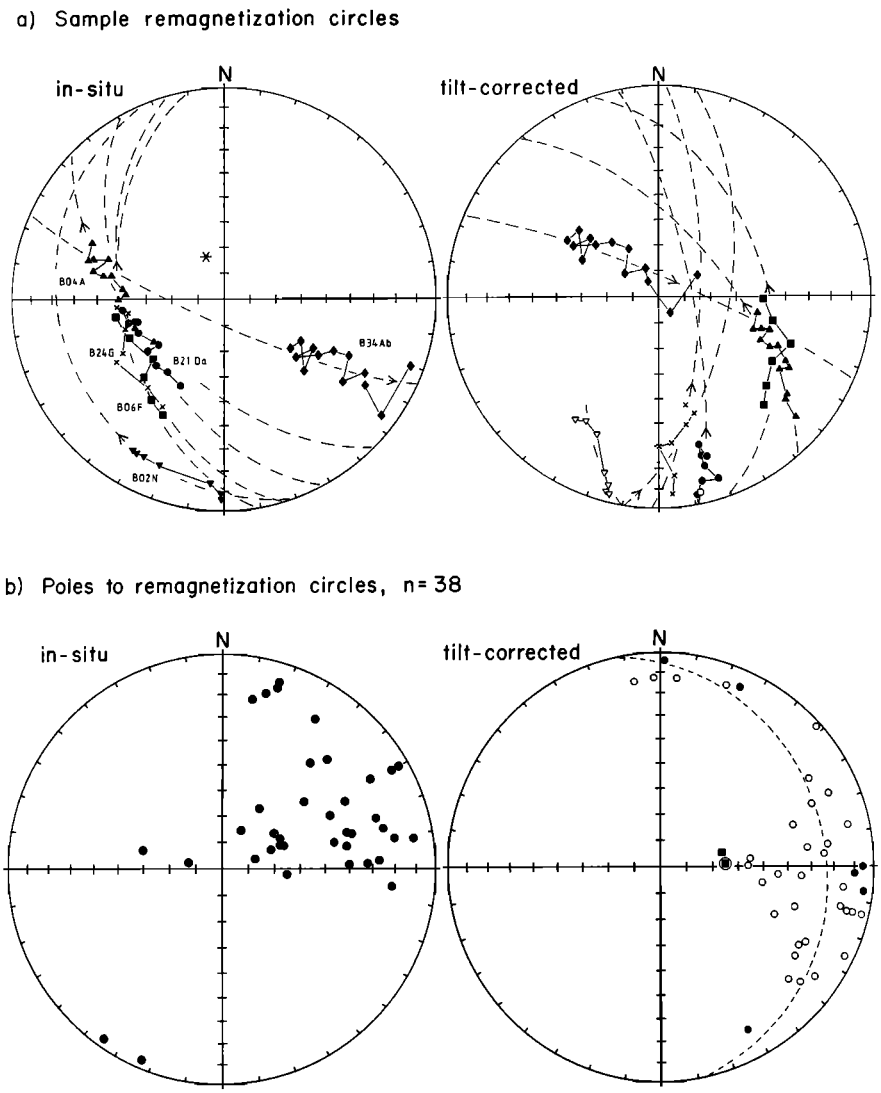


Fig. 10. (a) Equal-area projections showing successive magnetic directions during thermal demagnetization of selected sediment samples, in situ and after correction for tilt. The same symbols in both diagrams in (a) denote the same specimen. The lower-hemisphere remagnetization circle calculated for each sample is also shown; the range of temperatures and number of treatment steps, t, and the mean radius of the 95% error ellipse about the great-circle pole for each sample, [Bailey and Halls, 1984], are as follows -- B02N: 250° to 520°C, t=7, ε=7.4°; B04A: 260° to 605°C, t=13, ε=5.4°; B06F: 260° to 540°C, t=6, ε=11.7°; B21'Da: 380° to 605°C, t=9, ε=5.8°; B24G: 300° to 575°C, t=7, ε=12.6°; B34.Ab: 260° to 595°C, t=13, ε=10.7°. The trend for each individual great circle trajectory is indicated with an arrow. The present earth field (PEF) direction is marked by an asterisk in (a). Poles to sample remagnetization circles for sediment samples are shown in (b), plotted in situ and after two-stage tilt correction. The dashed line on the right indicates the best-fit upper hemisphere great circle to remagnetization circle poles and the filled square is the lower hemisphere pole to this great circle (D=77°,I=+66°); the circled square denotes the combined mean direction (D=87°,I=+65°) of all high-Tub components (Figure 9) and remagnetization circles for the sediments.

TABLE 4. F Ratio Values for Consistency Tests of Paleomagnetic Directions from the Bourinot Sediments and Volcanics

Remagnetization Circle Data for the High-Tub Component in the Sediments	Mean Decl/Incl	(1) Comparison With the Stable End Point Directions for the Sediments		(2) Comparison With the Stable End Point Direction[a] for the Volcanics	
		F_{calc}	F_{crit}	F_{calc}	F_{crit}
Simple T.C.					
Remagnetization circles only	241/-68	2.98[b]	3.19	5.30	3.25
Two-stage T.C.					
Remagnetization circles only	257/-66	1.85[c]	3.19	2.92	3.25
Combined remagnetization circles and stable end points	267/-65	...	...	2.26	3.18

Tub is laboratory unblocking temperature; Decl/Incl are declination and inclination, in degrees; T.C. is correction for tilt of the strata: "simple T.C." is a rotation along small circles about the strike direction, "two-stage T.C." is first a rotation along small circles to restore minor fold axes to horizontal then a second rotation about the new (unplunged) strike direction. In (1): F_{calc} is the value of the F ratio calculated following Bailey and Halls [1984], comparing the directions estimated from N=38 remagnetization circle poles and M=9 stable end points; F_{crit} is the critical value of F for 2 and 2M+N-4 degrees of freedom, at the 5% level of significance. In (2): F_{calc} is calculated for a comparison between the volcanic mean and the direction estimated from either N=38 remagnetization circles, or, in the case of the combined result, for N=38 remagnetization circles plus M=9 stable endpoint directions; F_{crit} is the critical value of F with 2 and 2M+N-2 degrees of freedom.

[a] The mean direction for the volcanics used in the comparison tests is Decl/Incl=293/-66.

[b] The mean stable end point directions for the sediments are Decl/Incl=276/-56 (simple T.C.)

[c] Decl/Incl=278/-59 (two-stage T.C.).

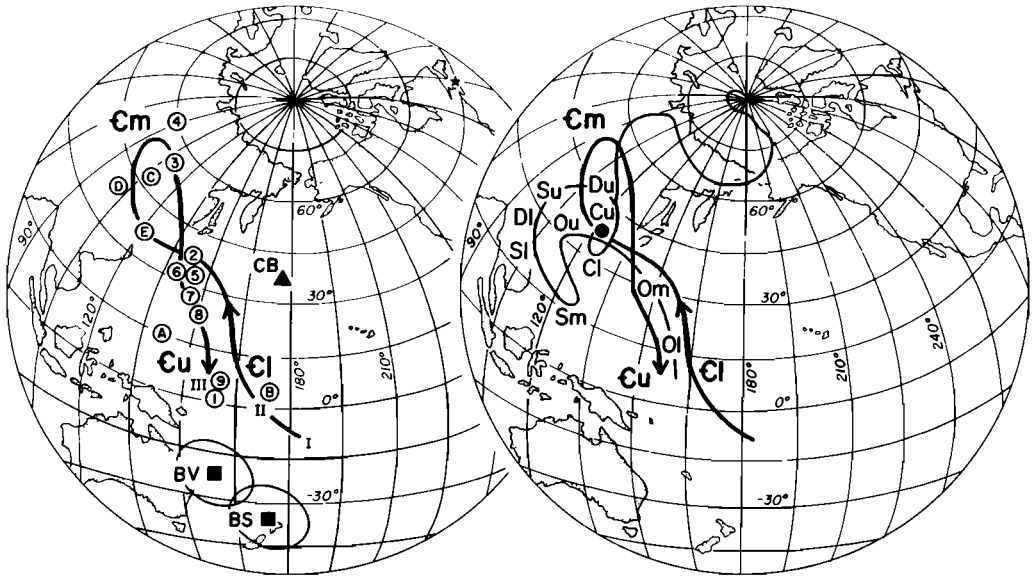


Fig. 11. The paleopoles for the Bourinot sediments (BS) and volcanics (BV) from Table 3, compared with Cambrian aged poles from the North American craton (open symbols). Pole CB (triangle) is from Rao et al. [1981]. The labels used for the open circles are the same as those used by Watts et al. [1980a]; the Roman numerals (I-III) denote poles from Dankers and Lapointe, [1981]. The sampling site is indicated on the left with a star. The heavy lined curve indicates the Cambrian loop, which is compared on the right with the North American APW curve for the later Phanerozoic [e.g., Van der Voo and Jackson, 1984]; the heavy solid dot is the pole obtained in this study for the Late Paleozoic overprint directions.

mean direction (D =87°, I=+65°; circled solid square of Figure 10b) and to test the consistency of these directions. Simple F ratio tests for consistency between remagnetization circle poles and the stable endpoint directions, or have been performed on the Bourinot data. The F ratios calculated for a comparison of (1), only the sediment data, and (2) the sediment data with the volcanic mean are given in Table 4. The table clearly shows that the difference between the remagnetization circle intersections after two-stage correction and after simple tilt correction is not great; nevertheless, the two-stage corrected poles show better consistency with both sediment stable end point directions and with the volcanic mean. The only clearly negative result, indicating statistically different means (at the 95% confidence level) is for the simple tilt-corrected remagnetization circle poles and the volcanic mean. We infer that the two-stage tilt correction for the sediments, based on field observations, is supported by the better agreement between the direction for the volcanics and the two-stage corrected data for the intercalated sediments.

In summary, these data for the Bourinot sediments show that a prefolding component of magnetization with steep inclination appears to be present in many of them. The direction obtained from the intersection of remagnetization circles is similar to that found for high-Tub components in a small number of sediment samples. The resultant direction, found by combining sample directions and remagnetization circles (see Table 3), is statistically indistinguishable from the direction for the apparent primary magnetization in the Bourinot volcanics.

CAMBRIAN PALEOLATITUDES OF NOVA SCOTIA AND CRATONIC NORTH AMERICA

A primary age for the characteristic magnetizations in the Bourinot volcanics has been argued on the basis of a fold test and the presence of dual-polarity magnetizations observed in separate unoxidized and oxidized flows. Because each volcanic flow occupies a slightly

different stratigraphic horizon, each apparently acquired a magnetization at a slightly different time, and their overall mean direction can be considered as giving an estimate of a time-averaged Mid-Cambrian field direction with respect to Nova Scotia. The similar direction obtained for the high-Tub magnetization of the sediments supports our contention that these directions are of Mid-Cambrian age. If we assume that the Mid-Cambrian field was dominantly geocentric and dipolar in its configuration, then we can use these paleomagnetic results to find Middle Cambrian pole positions relative to Nova Scotia and to infer its paleogeographic position at the time the Bourinot rocks acquired their magnetization.

Paleopole positions corresponding to the mean directions found for the Bourinot volcanics (D=293°, I=-66°) and sediments (D=267°, I=-65°) are given in Table 3, and are shown in Figure 11 where they can be compared with the Paleozoic apparent polar wander (APW) path for North America. Clearly the choice of polarity for the Bourinot results is arbitrary, but we have chosen the directions which yield the minimum discrepancy with respect to the normal-polarity APW curve for cratonic North America. The differences between the Bourinot paleopoles and the North American Cambrian poles in Figure 11 result for a major part from paleolatitude differences as discussed below, and no simple in situ rotation of the sampling locality will bring them into agreement. A paleopole corresponding to the magnetization found in a suite of Cambrian-age granites from Cape Breton Island (CB)[Rao et al., 1981] is also given in Figure 11 (triangle). As shown in the figure the CB pole is different from the poles for the Bourinot rocks as well as from the North American path. Given that there is considerable uncertainty in the age of the CB pole (obtained for plutons that have given radiometric ages spanning the range of 520±70 to 580±70 m.y.), the difference between it and the Bourinot poles may be a result of their quite different ages. Paleopoles from additional "younger" Paleozoic plutons from Cape Breton, not shown in Figure 11, have equally uncertain ages [Rao et al, 1981]. For this reason we concentrate on a comparison between the single results for the Bourinot rocks and the Cambrian data for cratonic North America.

A complication in this comparison is that the Cambrian APW path for the craton is problematic, as noted by many authors [Watts et al., 1980a, b; Dankers and Lapointe, 1981]; this path, determined by a sequence of poles from differently aged Cambrian strata, is characterized by a "Cambrian loop" [Watts et al., 1980a, b] which begins near 160°E, 0°N (Lower Cambrian) and trends toward progressively higher latitudes in the Mid-Cambrian, and then goes back on itself in the Upper Cambrian ending near the equator at 160°E at the end of the period (see Figure 11). If this path is real, it can be interpreted to represent an initial clockwise rotation of North America from Early to Middle Cambrian time, followed by a counterclockwise rotation occurring between Middle and Late Cambrian time [Watts, 1979]. However, because the higher-latitude Cambrian poles from the craton fall near to younger (Late Paleozoic) poles for North America (Figure 11), there exists the strong possibility that these mainly Mid-Cambrian age poles from the craton represent remagnetized directions, much younger than the age of the host strata. Indeed, probable Late-Paleozoic remagnetizations of Cambrian rocks from the craton have been suggested in a number of recent papers [Gillett, 1982; Wisniowiecki et al., 1983; Lynnes and Van der Voo, 1984]. Thus, at present the validity of the Cambrian loop has not been demonstrated conclusively.

Whatever the effect of this APW-loop on the rotation of North America, it is highly likely that this craton remained in equatorial paleolatitudes. Thus we can examine the paleolatitudes of cratonic North America and Cape Breton Island in a comparative plot showing the Cambrian paleolatitudes of the study area as observed and as hypothetically calculated from the cratonic APW path for our study area (Figure 12). The mean paleolatitude (Table 3) corresponding to the observed dual-polarity characteristic magnetizations of the Bourinot sediments and volcanics shows a significant difference with the expected paleolatitude for Nova Scotia if it had never moved with respect to the craton. This difference is in fact an estimate of the minimum relative separation during the Cambrian between Nova Scotia and cratonic North America. Figure 12 further shows that if the Mid-Cambrian APW loop for North America is valid, the minimum separation of Nova Scotia and North America is

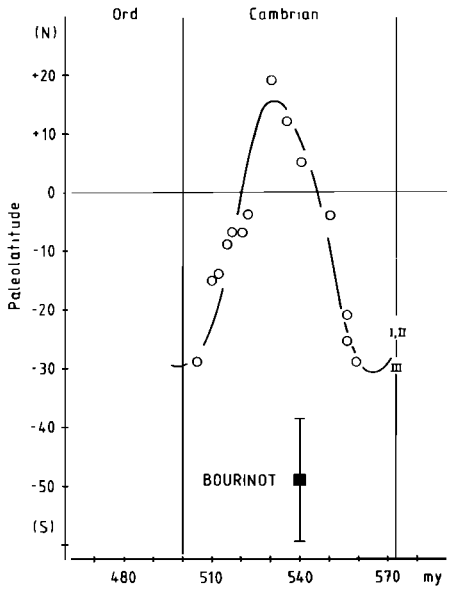


Fig. 12. The observed Mid-Cambrian paleolatitude for Nova Scotia corresponding to the mean inclination for the Bourinot volcanics and sediments, compared with paleolatitudes predicted for the study area from cratonic poles in Figure 11. The error bar for the Bourinot result indicates dp, see Table 3.

greater (~70°) than in the case where the Cambrian-loop is discounted, i.e., when taking only the predicted paleolatitudes that correspond to the Early and Late Cambrian cratonic poles. In the latter case, a still large discrepancy between the expected and observed paleolatitudes of about 20°±11° is inferred. Given the uncertainties in the Mid-Cambrian paleomagnetic data for North America, however, our estimate of the minimum separation ranges from ~20°±11° to 70°±11° km.

Reliable Cambrian paleomagnetic results for other continents are few in number, but some which are available indicate that the mid-Cambrian paleolatitude for the Avalon terrane is similar to that of Armorica and near that of Gondwana's north African margin [e.g., Hagstrum et al., 1980; Perigo et al, 1983]. Assuming the high Mid-Cambrian paleolatitude for Nova Scotia to be in the southern hemisphere, it is not illogical to assume that Gondwana, Armorica, and Nova Scotia were in proximity and drifting together.

CONCLUSIONS

Tilt-corrected magnetizations in volcanic units of the Bourinot Group with steep directions and dual polarities are probably primary, and they record the paleomagnetic field direction relative to Nova Scotia in the Middle-Cambrian. Remagnetization-circle analysis and stable end point directions reveal similar tilt-corrected directions in interbedded sedimentary units. The directions for the Bourinot rocks imply a high (49° S±11°) paleolatitude for Nova Scotia during the Mid-Cambrian, which is significantly different from the range of possible values for the predicted Mid-Cambrian paleolatitude of Nova Scotia (from 29°S to 18°N), inferred from the North American APW path. The discrepancy between predicted paleolatitudes and the observed paleolatitude implies that Nova Scotia was not part of North America in the Mid-Cambrian. They were presumably separated by an ocean and the difference in paleolatitudes (20°±11°) yields an estimate for the width of this ocean. The paleomagnetically determined position for Nova Scotia in the Mid-Cambrian thus provides evidence for the type locality of the "Wilson-cycle" in terms of the closing of an ocean basin during the Paleozoic, which has been suggested to explain the observed Early Paleozoic faunal differences between Avalon and North America.

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